EXP NO - 4
DATE -

TO IMPLEMENT PACKET SNIFFING AND TO WRITE A CODE FOR THAT USING RAW SOCKETS

AIM: To write a program using RAW sockets to capture and sniff network packets and display their basic information.

REQUIREMENTS:

* Python 3
* RAW socket support
* Linux / Ubuntu system
* Root / sudo privileges.

PROCEDURE:

1) Open the Terminal in Ubuntu / Linux.
2) Create a Python file:
   nano packet_sniffer.py
3) Write the RAW socket program to capture packets.
4) Save the file and exit.
5) Run the program with root privileges:
   Sudo python3 packet_sniffer.py.
6) The python program captures packets passing through the network interfaces.
7) It extracts and displays information such as:
   - Source IP address
   - Destination IP address
   - Protocol
   - Packet length.
8) Generate some network traffic by opening a website or using another terminal.
9) Observe the captured packet information
10) Stop the program using Ctrl+C.

PROGRAM:

```
import socket
import struct
s = socket.socket(socket.AF_INET, socket.
                  sock_RAW, socket.IPPROTO_IP)
while True:
packet, addr = s.recvfrom(65565)
ip_header = packet[0:20]
iph = struct.unpack('!BBHHHBBH4s4s',
                    ip_header)
 version = iph[0]>>4
  ih! = (iph[0] & 15)*4
  Protocol = iph[6]
source = socket.inet_ntoa(iph[8])
destination = socket.inet_ntoa(iph[9])
print("Source IP : ", source)
print("Destination IP : ", destination)
print("Protocol : ", protocol)
print("Packet Length : ", len(packet))
print("_"*40)
```

SAMPLE OUTPUT:

```
Source IP: 192.168.1.5
Destination IP: 142.250.183.4
Protocol: 6
Packet Length: 60

Source IP: 142.250.183.14
Destination IP: 192.168.1.5
Protocol: 6
Packet Length: 52
```

RESULT: Thus, a python program using RAW sockets was successfully implemented to capture and sniff network packets and display their source IP, destination IP, protocol, and packet length.